Introduction to Representation Theory

Motivation Given a group G Feb 7 Week 1-1

Q Can we find a group homomorphism

$$G \to GL_n(\mathbb{C}) \quad ?$$

E.g. Is there an injective one?

How do those homomorphisms look like?

Note: Such a homomorphism $G \to GL_n(\mathbb{C})$ gives an action of G on $\mathbb{C}^n$.

Conversely. Consider S_3 permutation group on $\{1,2,3\}$

$\rightsquigarrow$ Can define $S_3 \to GL(\mathbb{C}^3)$

$$\sigma \mapsto (e_j \mapsto e_{\sigma(j)})$$

$\uparrow$ standard basis.

Basic definitions

G = group (will usually assume that e.g. G is finite or G is compact)

K = field usually $\mathbb{R}$ or $\mathbb{C}$.

V = finite-dimensional K-vector space.

Defn A linear representation of G (over K) on V is a group homomorphism $\pi: G \to GL(V)$

- $\dim V$ = degree / dimension of π.

• V is called the representation space of π.

Will denote such a representation by π or (π, V).

Defn. A matrix representation of G over K is a group homomorphism $\pi: G \to GL_n(K)$ for some $n \in \mathbb{N}$

n = dimension of π.

Note Have canonical isomorphism $GL_n(K) \xrightarrow{\simeq} GL(K^n)$ by the standard basis

$\rightsquigarrow$ Can always view a matrix representation as a linear representation in a canonical way

<u>Defn</u> Suppose (π_1, V_1) and (π_2, V_2) are linear representations of G Then we call π_1, π_2 are equivalent (or isomorphic) written as $\pi_1 \simeq \pi_2$ If there is an isomorphism $\Phi: V_1 \to V_2$ s.t.

$$\forall g \in G \quad \pi_1(g) = \Phi^{-1} \pi_2(g) \Phi$$

Similarly if $\pi_1: G \to GL_{n_1}(K)$, $\pi_2: G \to GL_{n_2}(K)$ are matrix representations, then π_1, π_2 are called equivalent if $n_1 = n_2 =: n$ and $\exists A \in GL_n(K)$ s.t.

$$\forall g \in G. \quad \pi_1(g) = A^{-1} \pi_2(g) A.$$

<u>Note</u> Given a linear repn (π, V) of G and a basis $\mathcal{B}$ of V, can attach a matrix repr

$$\mathcal{B} = \{v_1, \dots, v_n\}$$

$$\sigma_{\mathcal{B}} : G \to GL_n(K)$$

by writing $K^n \xrightarrow{\simeq} V \qquad (x_j)_{j=1}^n \mapsto \sum x_j v_j$

Note: If $\mathcal{B}'$ is another basis of V, then $\sigma_{\mathcal{B}} \simeq \sigma_{\mathcal{B}'}$

Hence: Can attach an equivalence class of matrix representation to a given linear space.

In fact; Obtain a bijection

$$\{\text{equivalence classes of linear repns}\} \xleftrightarrow{\text{bij}} \{\text{eq classes of matrix repn's}\}$$

<u>Remark</u>: We are usually interested in representations up to equivalence, so we will often just talk about representations.

Remark: If (π, V) is a repn of G, H another group

$\varphi : H \to G$ a group homomorphism, then

$(\pi \circ \varphi, V)$ is a representation of H on V.

<u>Examples</u>: Representations of $(\mathbb{R}, +)$ over $K = \mathbb{R}$ or $\mathbb{C}$

1-dim rep'ns. $\pi : \mathbb{R} \to GL_1(K) = K^\times$

Given $\alpha \in K$, get a repn $\pi_\alpha : \mathbb{R} \to K^\times$

$$t \mapsto e^{\alpha t}$$

In fact if we assume that π is differentiable, those are all

$$\pi(t+s) = \pi(t)\pi(s)$$

$$\pi'(t) = \frac{d}{ds}\pi(t+s)\Big|_{s=0} = \pi'(0)\pi(t)$$

$\Rightarrow \pi(t) = \beta \cdot e^{\alpha t}$ for suitable $\beta, \alpha \in K$.

$\pi(0) = 1 \Rightarrow \beta = 1$.

This is wrong if we allow non-continuous representations

$$\pi : \mathbb{R} \to \mathbb{R}^\times$$

Fix a basis $\mathcal{Q}$ of $\mathbb{R}$ over $\mathbb{Q}$ wlog $1 \in \mathcal{Q}$

Define $\pi : \mathbb{R} \to \mathbb{R}^\times$ by $\pi(1) = e$ $\pi(q) = 1$ $\forall q \in \mathcal{Q} \setminus \{1\}$

Can extend to a group homomorphism $\pi : \mathbb{R} \to \mathbb{R}^\times$ but

$\pi \neq \pi_\alpha$, $\forall \alpha \in \mathbb{R}$

n-dim rep'n of $(\mathbb{R}, +)$ over $K = \mathbb{R}$ or $\mathbb{C}$

<u>Defn</u> If $A \in M_n(K)$, we define the matrix exponential e^A by

$$e^A = I_n + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$$

$$= \sum_{k=1}^{\infty} \frac{A^k}{k!}$$

<u>Proposition</u>: The series converges absolutely for every $A \in M_n(K)$. Moreover:

(i) $\forall A \in M_n(K)$, $e^A \in GL_n(K)$.

(ii) $e^{0_n} = I_n$.

(iii) $(e^A)^{tr} = e^{A^{tr}}$

(iv) $B^{-1} e^A B = e^{B^{-1}AB}$ $\forall A \in M_n(K)$, $B \in GL_n(K)$

(v) $\forall A, B \in M_n(K)$ s.t. $\underline{AB = BA}$ $e^{A+B} = e^A e^B$

In particular, $e^A e^{-A} = I_n$.

Now for $A \in M_n(K)$, define

$$\pi_A : \mathbb{R} \longrightarrow GL_n(K)$$
$$\pi_A(t) = e^{tA}$$

This is a group homomorphism by Prop (ii) & (iv)

note $(tA)(sA) = (sA)(tA)$

One can: If $\pi : \mathbb{R} \to GL_n(K)$ is a differentiable representation, $\exists A \in M_n(K)$ s.t $\pi = \pi_A$

<u>Example</u>: Rotations in $\mathbb{R}^2$

Consider rotations in $\mathbb{R}^2$ by some angle $t \in \mathbb{R}$.

$$\pi : \mathbb{R} \longrightarrow GL(\mathbb{R}^2) = GL_2(\mathbb{R}).$$

This is a group homomorphism, i.e. π is a 2-dim representation of $\mathbb{R}$.

Can we find $A \in M_2(\mathbb{R})$ s.t $\pi = \pi_A$?

<u>Claim</u> $\pi = \pi_A$ for $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

<u>Proof</u>: $A^2 = -I_2 \quad A^3 = -A \quad A^4 = I_n$

$$\rightsquigarrow e^{tA} = \sum_{(k=2m)} \frac{(tA)^{2m}}{(2m)!} + \sum_{(k=2m+1)} \frac{(tA)^{2m+1}}{(2m+1)!}$$

$$A^{2m} = (-1)^m I_2 \qquad A^{2m+1} = (-1)^m A$$

$$= \underbrace{\sum_{m \geq 0} \frac{(-1)^m t^{2m}}{(2m)!}}_{\cos t} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \underbrace{\sum_{m \geq 0} \frac{(-1)^m t^{2m+1}}{(2m+1)!}}_{\sin t} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \qquad \square$$

<u>Example</u>: S_n = permutation group on $\{1, 2, \cdots, n\}$

Write $E_{ij} \in M_n(\mathbb{C})$ for matrix units

$$(E_{ij})_{kl} = \begin{cases} 1 & (k,l) = (i,j) \\ 0 & \text{else} \end{cases}$$

Define: $\pi : S_n \longrightarrow GL_n(\mathbb{C})$

$$\sigma \longmapsto A_\sigma = E_{\sigma(1),1} + E_{\sigma(2),2} + \cdots + E_{\sigma(n),n}$$

Claim: π is a rep'n of S_n.

Clearly $\pi(id) = I_n$.

Note that: $E_{ij} E_{lk} = \begin{cases} E_{ik} & j = l \\ 0 & \text{else} \end{cases}$

Hence $\pi(\sigma)\pi(\tau) = A_\sigma A_\tau$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} E_{\sigma(i) i} E_{\tau(j) j}$$

$$= \begin{cases} E_{\sigma(i) j} & i = \tau(j) \\ 0 & \text{else} \end{cases}$$

$$= \sum_{j=1}^{n} E_{\sigma(\tau(j)) j}$$

$$= \pi(\sigma \circ \tau)$$

$\square$

Alternatively, can define $\pi : S_n \to GL_n(\mathbb{C}) = GL(\mathbb{C}^n)$

via standard basis

by $\pi(\sigma) e_j = e_{\sigma(j)}$

<u>Example</u>: Group actions Suppose we are given a group action of G on a finite set X

Recall that a group action of G is a map

$$G \times X \to X \qquad (g, x) \mapsto g.x$$

s.t. $(e, x) = x \qquad g.(h.x) = gh.x$

i.e. we have a group of homomorphisms

$$G \to S(X) = \text{group of bijections } X \to X$$

$\pi : G \longrightarrow GL(V_X) \quad g \mapsto \pi(g)$

V_X = $\mathbb{C}$-vector spaces with basis $e_x \quad x \in X$
$\in K^{|X|}$

Then $\pi(g) e_x = e_{g.x}$

As for $G = S_n$, $X = \{1, 2, \dots, n\}$ one shows that π is indeed a representation of G.

<u>Example</u>: Suppose G finite. Left / right action of G on itself

$$l : G \times G \longrightarrow G \quad (g, x) \mapsto gx$$

$$r : G \times G \longrightarrow G \quad (g, x) \mapsto xg^{-1}$$

Those actions give the <u>left (or right) regular representations</u>

$L : G \longrightarrow GL(K[G])$ $\qquad$ $K[G]$ = K-vector space of all functions $f : G \to K$

$R : G \longrightarrow GL(K[G])$

$L(g) f(x) = f(g^{-1}x)$ $\qquad$ left regular representation

$R(g) f(x) = f(xg)$ $\qquad$ right regular representation

<u>Remark</u> If X finite $K[X]$ = K-v.s of all fcts $f : X \to K$

Then $K[X] \simeq V_X$

$$f \mapsto \sum_{x \in X} f(x) e_x$$